Practice Problems Set-3

Problem 1

Let

$$f(x, y) = x^3 + y^3 - 3xy.$$

1. Find all critical points of f.
2. Determine whether each critical point is a local maximum, local minimum, or saddle point using the second derivative test.

Problem 2

Consider the function

$$f(x, y) = x^2 + y^2 + 4xy - 6x - 6y.$$

1. Find the stationary points of f.
2. Classify the stationary points using the Hessian matrix.

Problem 3

Let

$$f(x, y) = x^4 + y^4 - 4xy.$$

1. Compute all critical points.
2. Use the Hessian determinant to determine the nature of each critical point.

Problem 4

Use the Taylor expansion of e^x about $x = 0$ to approximate $e^{0.01}$.
Assuming

$$e \approx 2.71828,$$

compute an approximation of $e^{0.01}$ up to the second order term.

Problem 5

Let $x, s \in \mathbb{R}^n$, A be an $n \times n$ matrix and W be a symmetric matrix. Prove that

$$\frac{\partial}{\partial s}(x - As)^T W(x - As) = -2(x - As)^T WA.$$

Problem 6

Let $f(x) = x^T Bx$, where B is an $n \times n$ matrix constant matrix and $x \in \mathbb{R}^n$.

1. Expand $x^T Bx$ in component form.
2. Differentiate with respect to x using partial derivatives.
3. Deduce the matrix identity

$$\frac{\partial}{\partial x}(x^T Bx) = x^T(B + B^T).$$